Analysis Research and analysis of neural network Internet economy under big data modeling - the Belt and Road Initiative as the background

Yuan Zhang*
Sichuan College of Arts and Sciences
Dazhou, Sichuan, China
383497659@qq.com

Abstract

Since the 18th National Congress of the Communist Party of China, the President of China has personally arranged the issues related to the Belt and Road Initiative, enhanced the ties between China and other countries, and further promoted the in-depth development of the community of shared future for mankind. The Belt and Road is short for the Silk Road Economic Belt and the 21st Century Maritime Silk Road. It is an international cooperation platform that adheres to multilateralism and an open and inclusive economic cooperation initiative provided by China to the world. Its historical origin can be traced back to the ancient Silk Road. In this context, the economic and trade exchanges between China and countries along the Belt and Road are closer, further promoting the connection between China and countries in Africa, Oceania, Asia, North America, South America, Europe and other countries.

CCS Concepts

• **Theory of computation** → Theory and algorithms for application domains; Database theory; Data structures and algorithms for data management.

Keywords

Internet economy, Big data, Mathematical modeling, the Belt and Road

ACM Reference Format:
Yuan Zhang. 2024. Analysis Research and analysis of neural network Internet economy under big data modeling - the Belt and Road Initiative as the background. In *2024 5th International Conference on Computer Science and Management Technology (ICCSMT 2024), October 18–20, 2024, Xiamen, China.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3708036.3708040

1 INTRODUCTION

As early as two thousand years ago, people began to trade. There was no currency in circulation, so people could only trade goods through barter, which accidentally promoted the interconnection between countries and further promoted the spread of culture.

In 2013, Chinese leaders first put forward the idea of jointly building the Belt and Road Silk Economic Belt. We will further promote common prosperity and peaceful development for all mankind and build more fair, equal, mutual assistance and friendly international relations through these measures.

In recent years, the global economic development has been under downward pressure, and import and export trade has been under pressure from multiple factors such as epidemic lockdowns, supply chain crises, geopolitical disputes and regional protectionism. However, due to the high cost of anti-globalization, the prospect of goods and services traded by countries and regions around the world based on comparative advantages is still promising [1-2] At the same time, limited offline channels during the epidemic also accelerated the popularity of online transactions, further promoting the penetration rate of global e-commerce. With the launch of the Belt and Road Silk Road Economic Belt, the upgrading of industrial structure of various industries, policy encouragement and the deepening of technological progress, China's cross-border export e-commerce has ushered in new development opportunities. [3-4]

Table 1: Factors affecting trade.

B	Influencing factors
Z1	Geography
Z2	The National economy
Z3	National consumption level
Z4	Trade routes
Z5	Import and export policy
Z6	Nature of commodity

2 ECONOMIC DATA MODELING

2.1 Logical model modeling and analysis

Taking the trade with African countries as the case material, this paper summarizes the factors affecting the trade (namely, region, national economy, national consumption level, trade transportation routes, import and export policies, and commodity nature) (this paper only lists the common factors studied, but does not represent all the factors), and then codes these factors, as shown in Table 1.

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ICCSMT 2024, October 18–20, 2024, Xiamen, China
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0999-9/24/10
https://doi.org/10.1145/3708036.3708040

Table 2: Assignment table of judgment matrix.

Z	Z1	Z2	Z3	Z4	Z5	Z6
Z1	1.00	0.70	0.46	0.66	0.94	0.40
Z2	1.43	1.00	0.59	0.26	1.51	0.31
Z3	2.17	1.69	1.00	1.52	2.42	0.76
Z4	1.52	3.85	0.66	1.00	2.40	0.85
Z5	1.06	0.66	0.41	0.42	1.00	0.28
Z6	2.50	3.23	1.32	1.18	3.57	1.00

2.2 Determination of the weight of single-level indicators

The eigenvector W corresponding to the maximum eigenvalue of the judgment matrix Z is obtained by using the sum method, which is the weight vector of the relative importance of the corresponding index at the same level to an index at the previous level after normalization. The specific steps are as follows (Table 2):

Converting the table to matrix Z.

$$Z = \begin{pmatrix} 1.00 & 0.70 & 0.46 & 0.66 & 0.94 & 0.40 \\ 1.43 & 1.00 & 0.59 & 0.26 & 1.51 & 0.31 \\ 2.17 & 1.69 & 1.00 & 1.52 & 2.42 & 0.76 \\ 1.52 & 3.85 & 0.66 & 1.00 & 2.40 & 0.85 \\ 1.06 & 0.66 & 0.41 & 0.42 & 1.00 & 0.28 \\ 2.50 & 3.23 & 1.32 & 1.18 & 3.57 & 1.00 \end{pmatrix} \tag{1}$$

$$C_{ij} = \frac{Z_{ij}}{\sum_{i=1}^{n} Z_{ij}} = \begin{pmatrix} 0.10 & 0.06 & 0.10 & 0.12 & 0.08 & 0.10 \\ 0.14 & 0.09 & 0.12 & 0.05 & 0.12 & 0.08 \\ 0.21 & 0.14 & 0.21 & 0.29 & 0.20 & 0.20 \\ 0.15 & 0.33 & 0.14 & 0.19 & 0.19 & 0.22 \\ 0.10 & 0.06 & 0.09 & 0.08 & 0.08 & 0.07 \\ 0.24 & 0.28 & 0.28 & 0.22 & 0.29 & 0.26 \end{pmatrix} \tag{2}$$

$$\overline{C_i} = \sum_{j=1}^{n} C_{ij} = \begin{pmatrix} 0.70 & 1.43 & 1.44 & 0.59 & 1.79 & 0.38 \end{pmatrix} \tag{3}$$

$$W_i = \frac{\overline{C_i}}{\sum_{i=1}^{n} \overline{C_i}} = \begin{pmatrix} 0.10 & 0.20 & 0.21 & 0.08 & 0.26 & 0.05 \end{pmatrix} \tag{4}$$

$$ZW_i = \begin{pmatrix} 0.70 & 0.46 & 0.66 & 0.94 & 0.40 & 1.76 \\ 1.00 & 0.59 & 0.26 & 1.51 & 0.31 & 1.75 \\ 1.69 & 1.00 & 1.52 & 2.42 & 0.76 & 3.25 \\ 3.85 & 0.66 & 1.00 & 2.40 & 0.85 & 3.89 \\ 0.66 & 0.41 & 0.42 & 1.00 & 0.28 & 1.97 \\ 3.23 & 1.32 & 1.18 & 3.57 & 1.00 & 3.83 \end{pmatrix} \times \begin{pmatrix} 0.09 \\ 0.20 \\ 0.21 \\ 0.08 \\ 0.26 \\ 0.05 \end{pmatrix} = \begin{pmatrix} 0.67 \\ 0.71 \\ 1.47 \\ 1.50 \\ 0.60 \\ 0.39 \end{pmatrix} \tag{5}$$

$$\lambda_{mac} = \frac{\sum_{i=1}^{4} \frac{ZW_i}{W_i}}{n} = 5.31$$

It follows from steps 1, 2, 3, 4 and 5, It can be concluded that Ci=0.012, and further according to the definition of consistency test, C.R=0.02<0.1, indicating that the matrix passes the consistency test.

3 ANALYSIS OF TRADE PATH

The export trade between China and countries along the Belt and Road has developed rapidly, and the trade routes are diversified. From the perspective of regional distribution, China has close trade contacts with countries along the Belt and Road in Asia, Europe, Africa and other regions. Among them, ASEAN is an important trading partner of China along the Belt and Road, [5]for which many central enterprises cooperate with countries along the Belt and Road to build and transform local transportation, including countries in Southeast Asia, Africa and Central Asia, providing relatively convenient basic hardware for facilitating economic and trade exchanges.

The Belt and Road Initiative is an important link in China's international trade layout. Early trade was mainly through Middle Eastern countries or Russia, but now it has a network of more than 50 countries. At the same time, in order to improve trade efficiency, China has also unified the layout of shipping, air transportation, railway transportation and other modes of transportation, and strengthened the construction of ports, electricity and roads.

In the context of the Belt and Road Initiative, more than 20 Chinese enterprises have also actively participated in the construction of other countries, mainly in the fields of bridge construction, basic power construction and construction construction, creating new drivers for the overall layout. [5-6]

It should be noted that specific trade routes may change and expand with time, changes in market demand, policy adjustments, and economic development of each country. Through data modeling and big data statistical analysis, it is found that the routes between China and the countries along the Belt and Road are very similar to neural networks, so we carry out analogy repetition and analysis, and obtain Figure 1.

As can be seen from Figure I, the Neural network distribution map is one of the computer model methods, is a computer model with the common characteristics of linear and nonlinear, in the application, usually used for program, logic, line and other aspects of the study, is a very intuitive method to study the model. The neural network used in this paper is somewhat different from the real neural network diagram. The neural network diagram in this study is composed by simulating the structure of the Belt and Road trade routes. [7-8] By inserting relevant codes into the model, the computer background will automatically simulate the optimal route, update and adjust the neural network itself, and timely update the

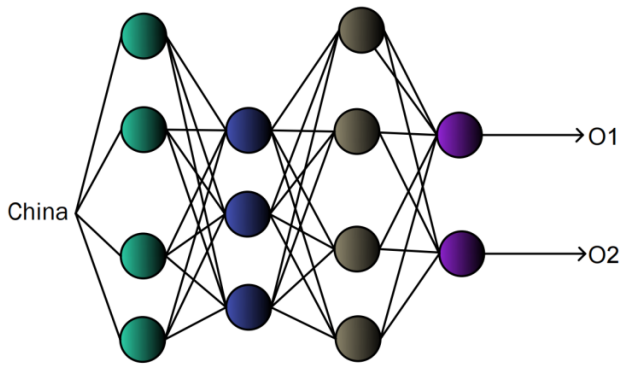

Figure 1: Neural network diagram of trade exports.

optimal choice between each trade port, including tariff, transportation time, stay time, sea, land and air transportation mode, etc., and provide the optimal route through comprehensive comparison. This can greatly reduce transportation costs and save transportation time.

4 ECONOMIC AND TRADE ANALYSIS OF THE BELT AND ROAD IN THE PAST FIVE YEARS

In 2022, China's net OFDI was US $163.12 billion, down 8.8 percent from the previous year. Among them:

New equity investment amounted to USD61.13 billion or 37.5%, reinvestment of current proceeds amounted to USD80.38 billion or 49.3%, and new investment in debt instruments amounted to USD21.61 billion or 13.2%. By the end of 2022, 29,000 Chinese domestic investors had set up FDI enterprises outside China (border)(see table 3)①, and 4.66 million enterprises were distributed in 190 countries (regions) around the world. [9]② The total assets of overseas enterprises at the end of the year were 84 trillion US dollars, and the cumulative net OFDI was 2,744.81 billion US dollars, including 1,484.56 billion US dollars of equity investment. Accounting for 53.9%, the proceeds were reinvested in US $962.64 billion, accounting for 34.9%, and debt instruments were invested in US $307.61 billion, accounting for 11.2%.

It can be seen from the above that the total amount of China's foreign investment is large, especially since the implementation of the Belt and Road Initiative, China's foreign investment has a significant impact.

At the same time, based on the big data of Tajikistan, Algeria, Russia, Cuba and other countries' import, export, total amount and freight container throughput to China in the past five years, we explore the trend and actual situation of domestic and foreign export trade under the background of the Belt and Road. Figure 2, Figure 3, Figure 4 and Figure 5 are obtained by research and analysis of relevant data.

Through Figure 2, 3, 4 and 5, it is found that under the background of the Belt and Road Initiative, we take the big data of Tajikistan, Algeria, Russia, Cuba and other countries' import, export, total amount and freight container throughput to China in the past five years as the research foundation to deeply explore the trend and reality of domestic and foreign export trade.[10-11]

Through a detailed analysis of these data, we find that China's trade with these countries shows a diversified and growing trend. In terms of imports, the input of various resources and featured products meets the different needs of the domestic market and promotes the optimization and upgrading of the industry. In the export sector, the market share of high-quality goods made in China has gradually expanded in these countries, covering a wide range of sectors from mechanical and electrical products to light industrial products, highlighting the strong strength of China's manufacturing industry.

The steady increase in total trade reflects the expanding depth and breadth of economic cooperation between the two sides. The change of freight container throughput intuitively reflects the development of efficiency and scale of logistics transportation.

In general, the Belt and Road Initiative has provided broader space and opportunities for trade cooperation between China and Tajikistan, Algeria, Russia, Cuba and other countries. In the future, with the continuous optimization of policies, continuous improvement of infrastructure and active participation of enterprises from both sides, trade exchanges are expected to be further strengthened to achieve mutual benefit and win-win results and jointly promote economic prosperity and development.

5 SUMMARY

Through big data analysis, we found the trade trend between China and other countries under the background of the Belt and Road, but there are still many external factors, the following are some suggestions;

Table 3: Classified composition of China's OFDI flows and stocks in 2022.

Classification	Financial flow		Stock of finance	
	Amount/US $100 million	Proportion	Amount/US $100 million	Proportion
Summation	1631.2	100	27548.1	100
Financial sector	221.2	13.6	3039	11
Non-financial	1410	86.4	24509.1	89

Note: 1. Financial refers to the investment that domestic investors invest directly in overseas financial enterprises multiple non-financial refers to the investment that domestic investors invest directly in overseas non-financial enterprises. 2. The difference between the non-financial flow data in 2022 and the express data of the Ministry of Commerce in 2022 (US $116.85 billion) is mainly due to the reinvestment of earnings.

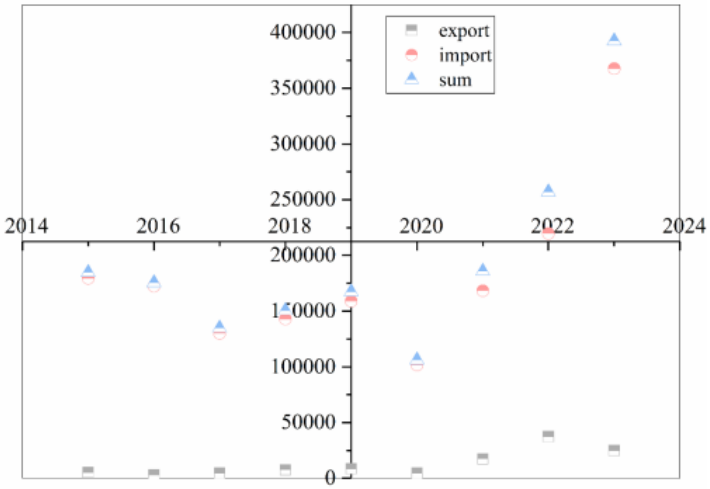

Figure 2: Tajikistan's imports, exports, and total imports and exports to China.

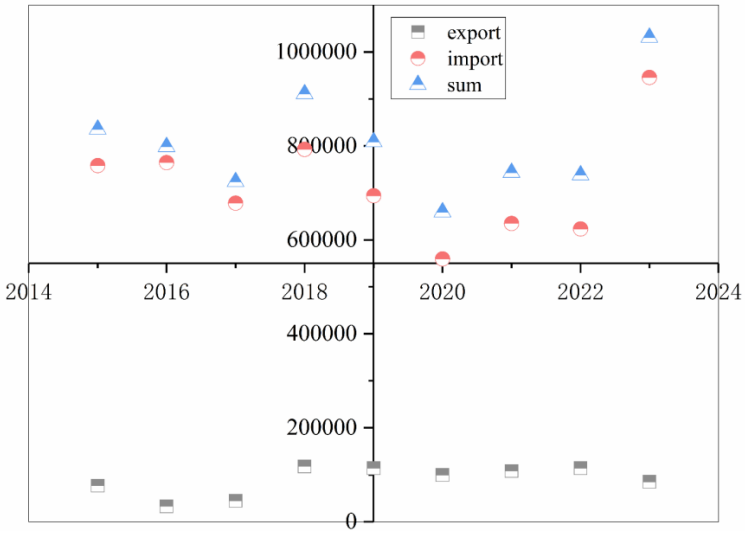

Figure 3: Algeria's imports to China, exports, and total imports and exports with China.

- Strengthen international cooperation on the Belt and Road Initiative. We should give full play to the role of multilateral and bilateral economic and trade cooperation mechanisms, take countries related to the Belt and Road Initiative as key sources of imports, strengthen strategic synergy, moderately increase the import of featured and high-quality products to meet the needs of upgrading domestic consumption, and expand trade scale.
- Strengthen diplomatic cooperation. We should continue to deepen exchanges between countries, elevate relations to an all-weather partnership or strategic partnership, promote people-to-people exchanges among countries, provide more appropriate policies for trade, and reduce trade costs caused by policies.
- Hold various trade fairs. Actively cooperate with other countries to hold strategic investment and trade promotion meetings in emerging industries, constantly strengthen exchanges between entrepreneurs of various countries, provide more diversified opportunities for cooperation between enterprises, and further promote the trading volume of trade cooperation and the volume of currency circulation.

References

[1] Liu Z, Ding Y. Enhancing China's image in Africa: The role of the Belt and Road Initiative[J]. China Economic Review, 2024, 87102239-102239.
[2] Li M. Research on the Effect of China's Direct Investment on Countries along the Belt and Road Based on RMB Internationalization[J]. Asian Journal of Economics, Business and Accounting, 2024, 24(7):215-225.
[3] International Institute for Strategic Studies: China's Belt and Road Initiative. A Geopolitical and Geo-economic Assessment. London: IISS (Strategic Dossier),

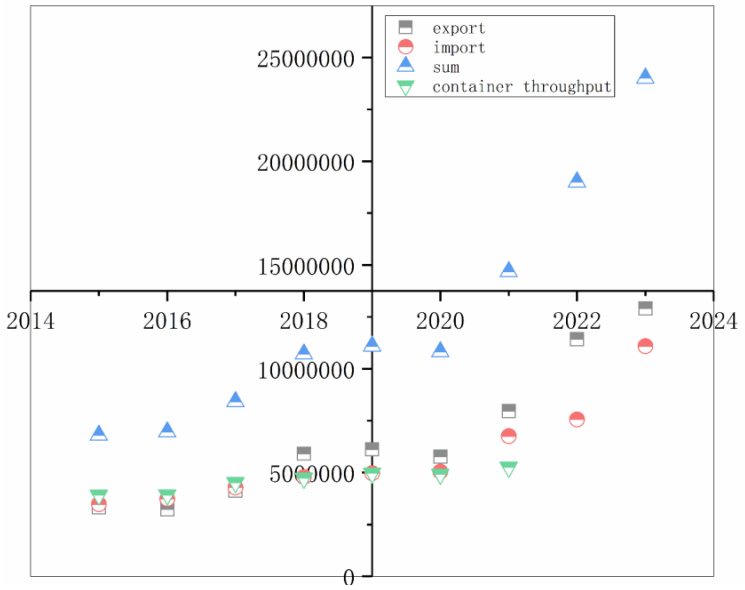

Figure 4: Russia's imports and exports to China, China's total imports and exports, and container throughput.

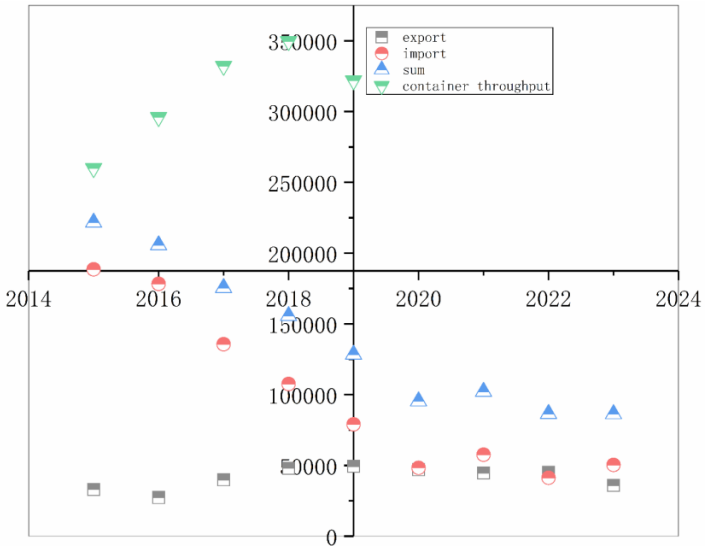

Figure 5: Cuba's imports and exports to China, China's total imports and exports, and container throughput.

November 2022[J]. SIRIUS-Zeitschrift für Strategische Analysen, 2024, 8(2):229-231.

[4] Islam S M, Tripura J U, Islam S M, *et al.* Appraising water resources for irrigation and spatial analysis based on fuzzy logic model in the tribal-prone areas of Bangladesh.[J]. Environmental monitoring and assessment, 2024, 196(7):641-641.

[5] Howick J, Weston B A, Solomon J, *et al.* How does communication affect patient safety? Protocol for a systematic review and logic model.[J]. BMJ open, 2024, 14(5):e085312-e085312.

[6] WanYi L, YunWen F, Da T, *et al.* Fault logic and data-driven model for operation reliability analysis of the flap deflection angle.[J]. Philosophical transactions. Series A, Mathematical, physical, and engineering sciences, 2023, 381(2260):20220385-20220385.

[7] Väänänen J, Welch D P. When cardinals determine the power set: inner models and Härtig quantifier logic[J]. Mathematical Logic Quarterly, 2023, 69(4):460-471.

[8] Tiase L V, Sward A K, Facelli C J. A Scalable and Extensible Logical Data Model of Electronic Health Record Audit Logs for Temporal Data Mining (RNteract): Model Conceptualization and Formulation.[J]. JMIR nursing, 2024, 7e55793.

[9] Park G, Lee Y, Cho M. Enhancing healthcare process analysis through object-centric process mining: Transforming OMOP common data models into object-centric event logs.[J]. Journal of biomedical informatics, 2024, 156104682.

[10] Huihui S, Qiang L. Robust variable selection in semiparametric mixed effects longitudinal data models[J]. Communications in Statistics - Theory and Methods, 2024, 53(3):1049-1064.

[11] ang W J. Analysis of persistence rate of cognitive enhancers in Alzheimer's disease using the common data model[J]. Alzheimer's & Dementia, 2023, 19(S22).